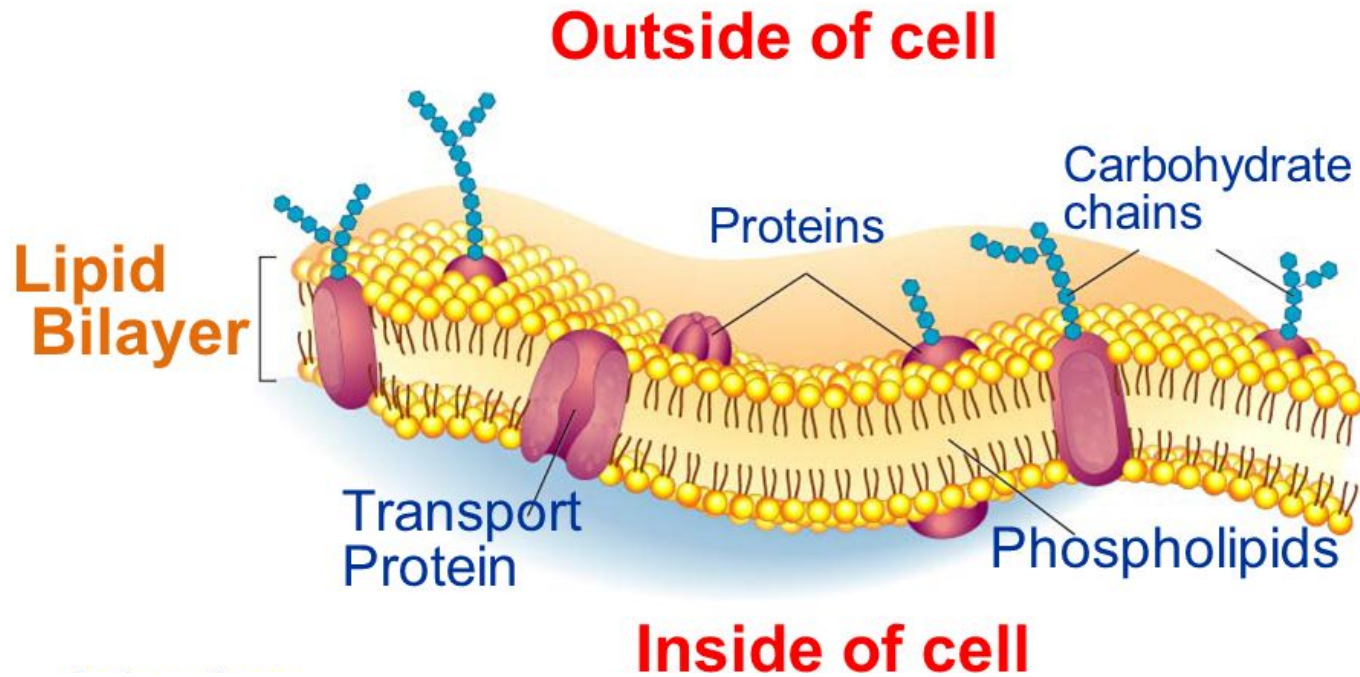


Lipids

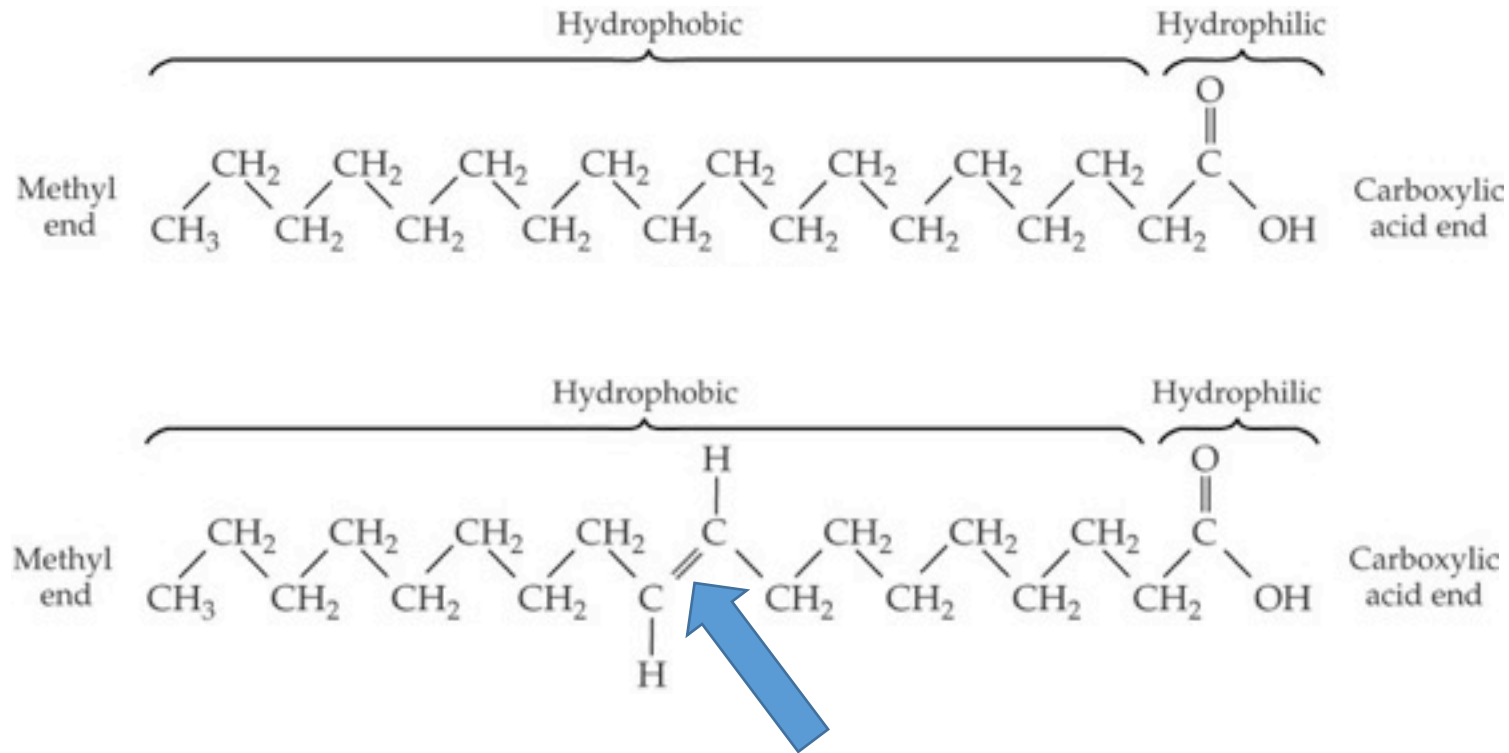


Dr. P. Satpati,
BSBE, IIT Guwahati

Lipids are of Structural or Biological Importance

- 1. Fatty Acids**
- 2. TAG (Triacylglycerides)/TG**
- 3. Steroid**

1. Fatty Acids



Saturated Fatty acid (SFA)

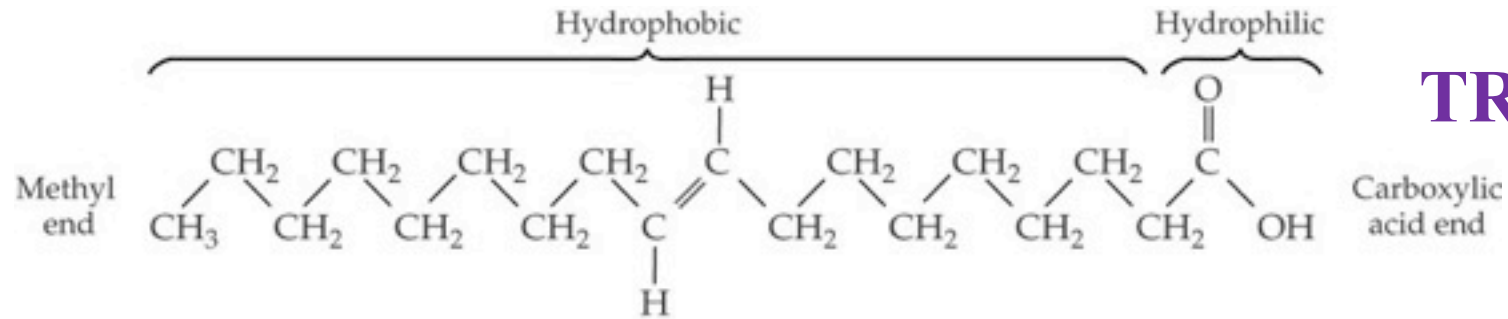
Mono-Unsaturated Fatty acid (MUFA)

Poly-Unsaturated Fatty acid (PUFA)

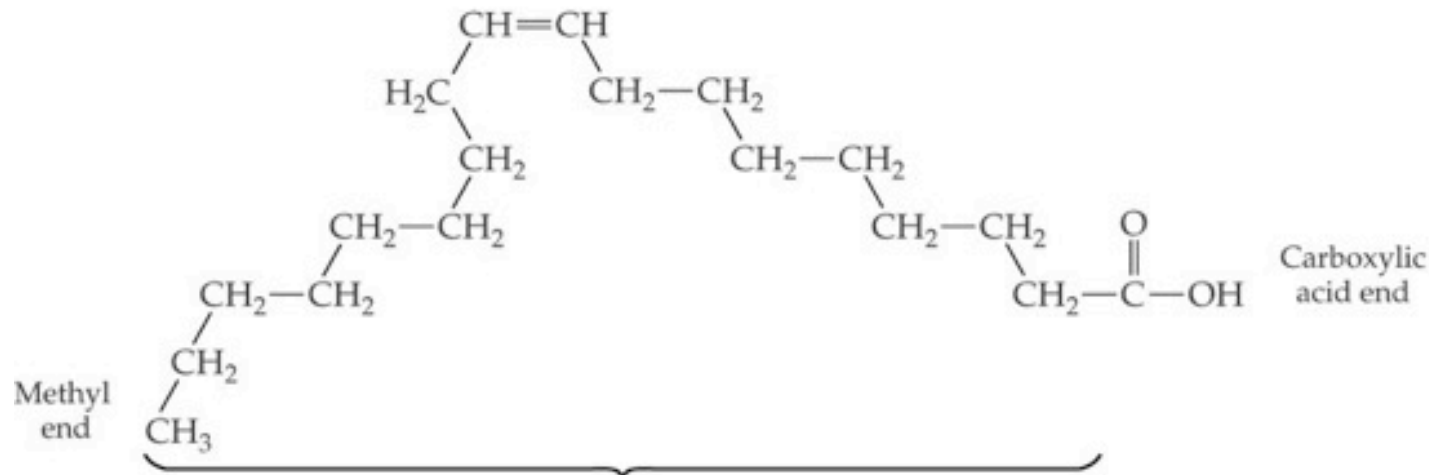
Fact: Most naturally occurring fatty acids have an unbranched chain of an even number of carbon atoms, from 4 to 28.

Fatty Acids

Double bond (Cis, Trans Configurations)



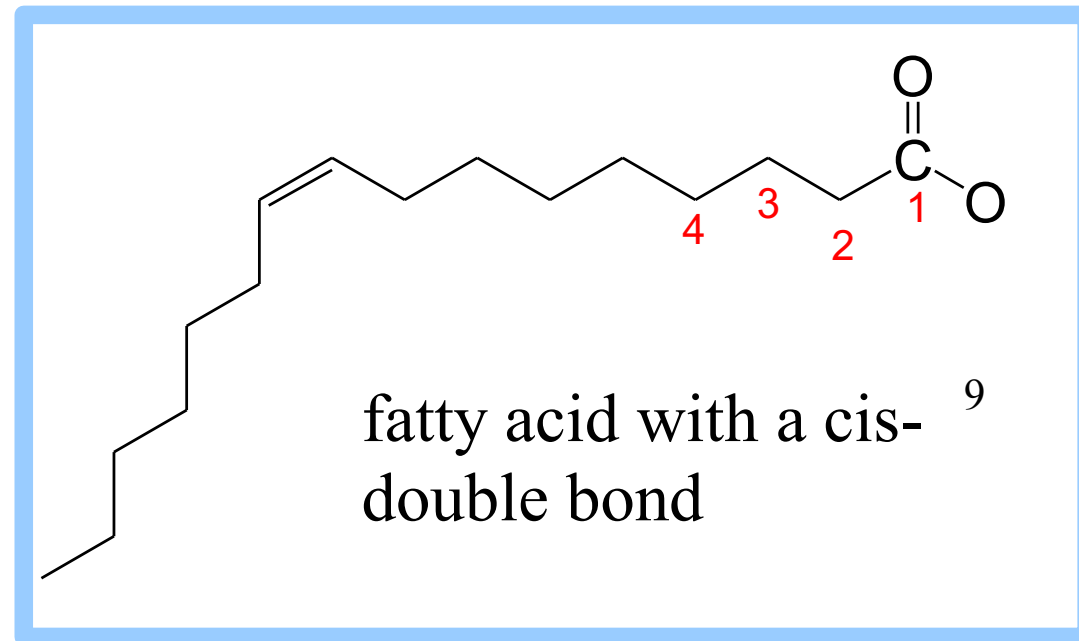
TRANS (Linear like Saturated FA)



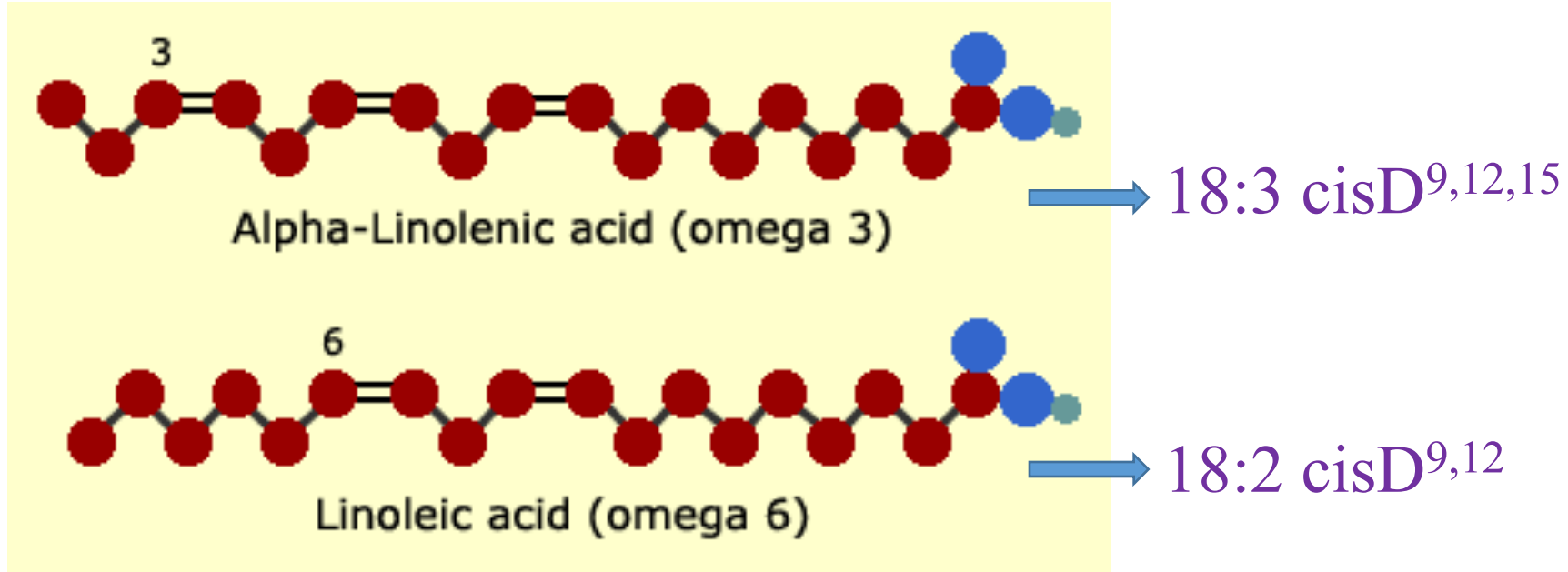
CIS (BENT, Produces KINK)

Nomenclature of fatty acids

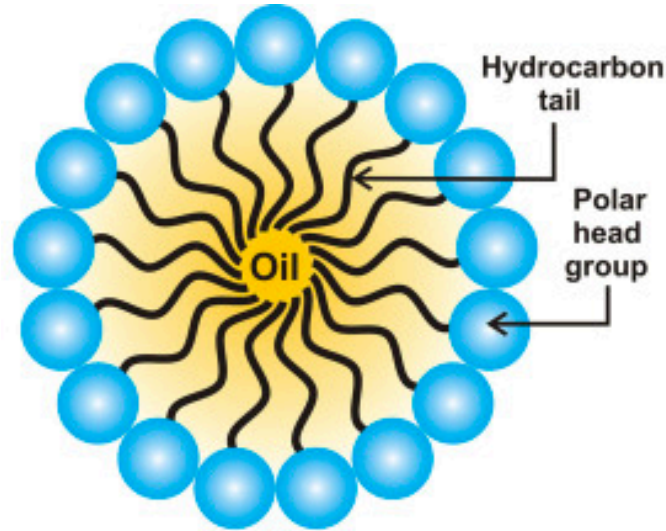
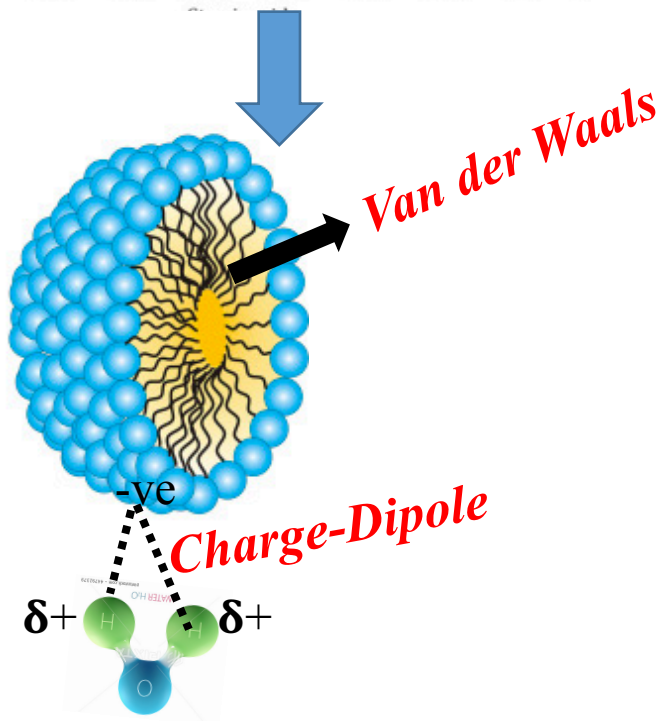
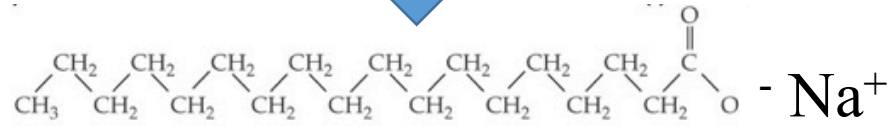
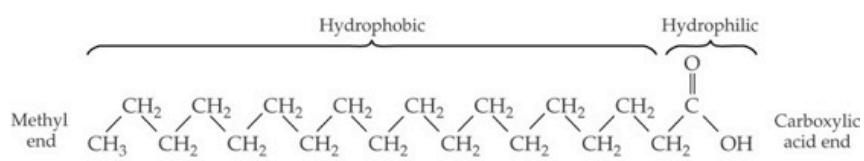
16-C fatty acid with one cis double bond between C atoms 9-10 : **16:1 cis D⁹**.



Two Essential fatty acids (Lecture 2, Essential nutrients)

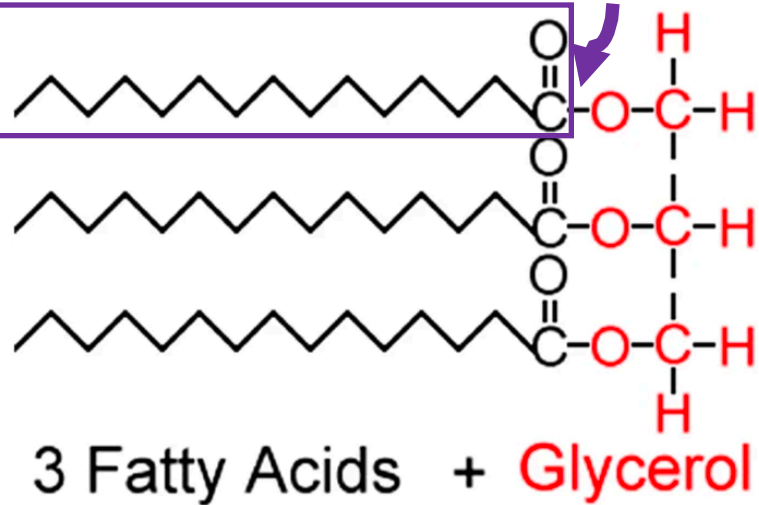


fatty acids and SOAP



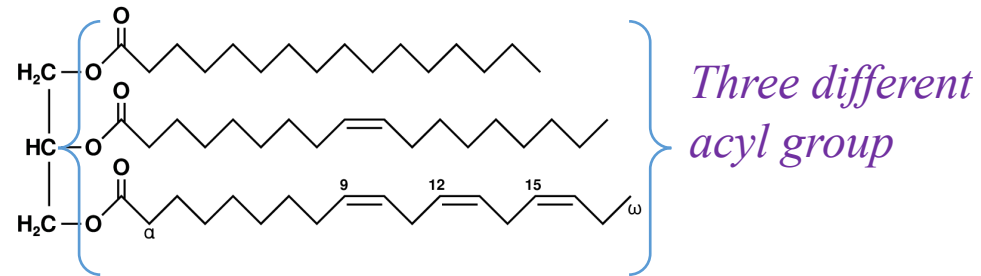
SOAP trap dirt/oil at the interior of its micelle.
Washing after applying soap take away the dirt as a micelle.

2. TAGs (Triacylglycerides)



TAGs :

- ❖ Three ester bonds
- ❖ Differ due to the type of attached acyl group

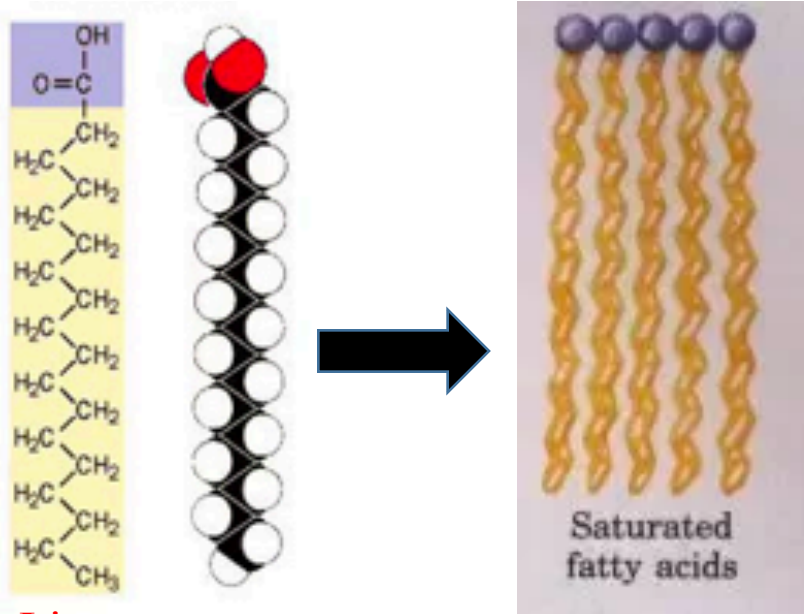


- ❖ If SOLID at room-temp → FAT
- ❖ If LIQUID at room-temp → Oil



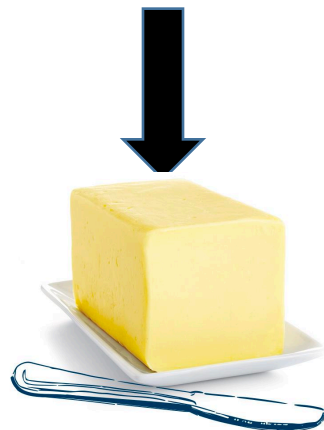
Solid/ Liquid state depends on the configuration (CIS/TRANS) of the Fatty acid

Saturated Fatty acid



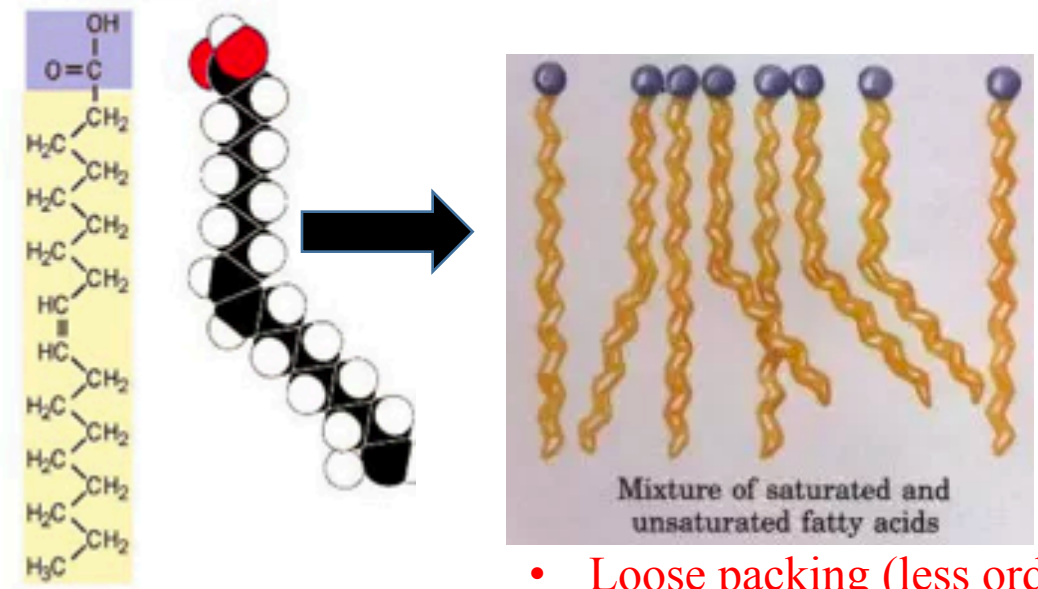
Liner

- Nice packing (well ordered)
- Maximum van der Waals attraction



Solid at
room Temperature

Unsaturated (CIS) Fatty acid



Kink

- Loose packing (less ordered)

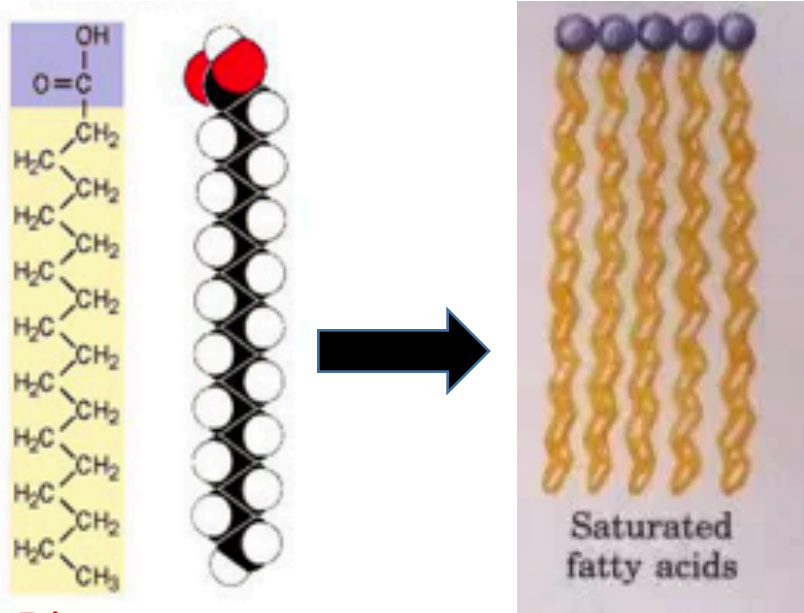


Liquid at room Temperature

Solid/ Liquid state depends on the configuration (CIS/TRANS) of the Fatty acid

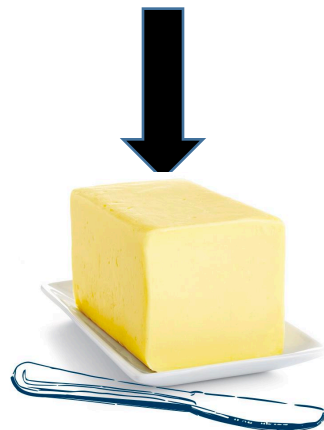
Also Unsaturated (TRANS) Fatty acid

Saturated Fatty acid



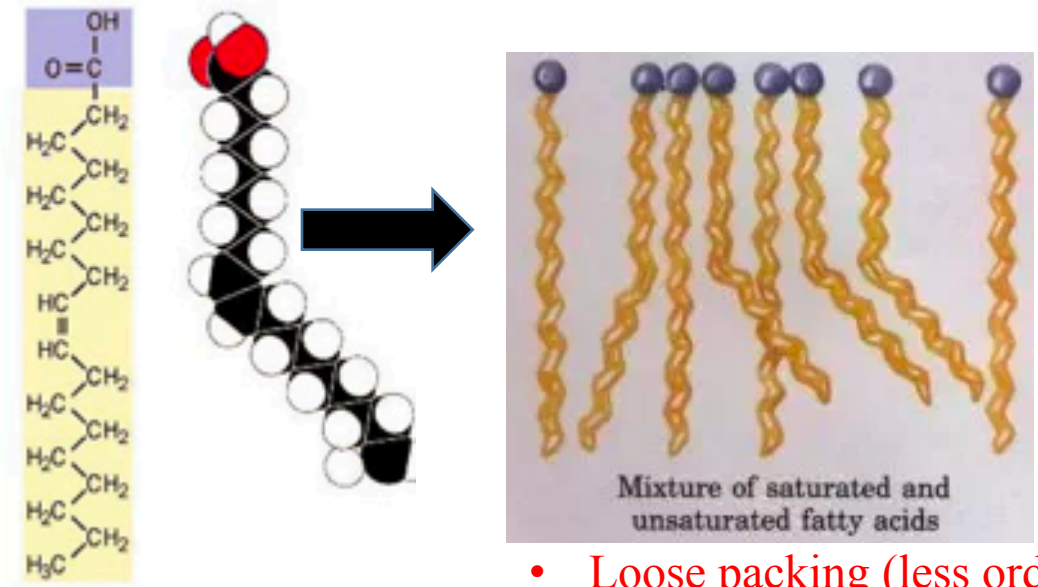
Liner

- Nice packing (well ordered)
- Maximum van der Waals attraction



Solid at
room Temperature

Unsaturated (CIS) Fatty acid



Kink

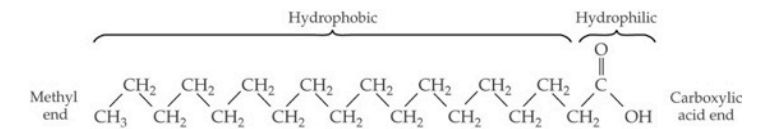
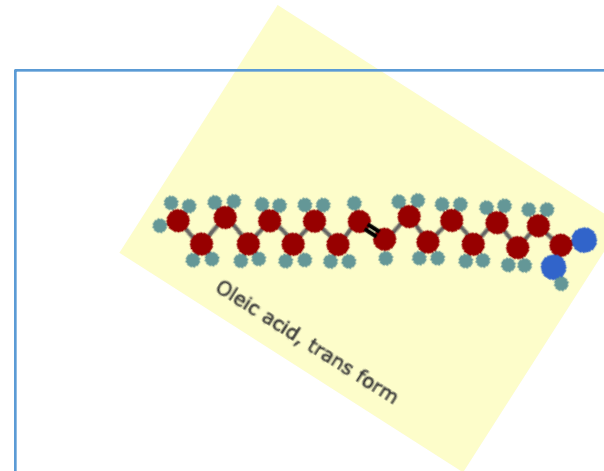
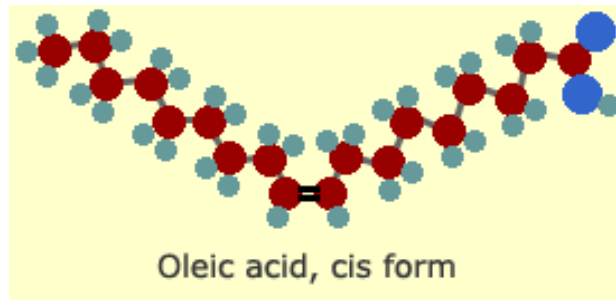
- Loose packing (less ordered)



Liquid at room Temperature

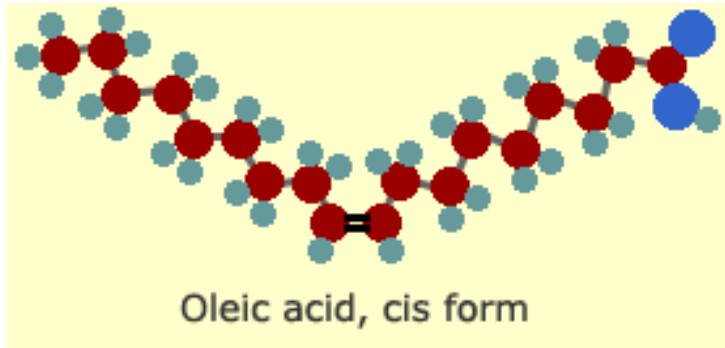
Properties of fatty acids

- Naturally occurring fatty acids have even number of carbon atoms (from 4 to 28).
- Saturated fatty acids/ Unsaturated Trans fatty acids are solid at room temperature.



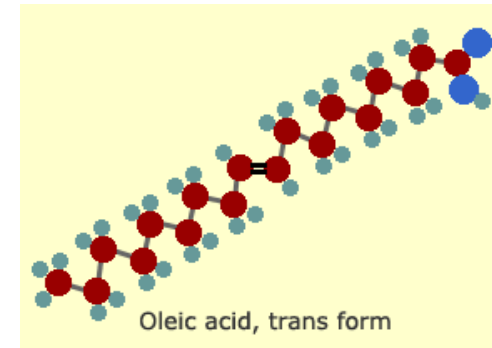
Cis double bonds disrupts intermolecular packing and therefore lower melting temperature.

Food industry: Adulteration of food



Deep fat frying →

Partially hydrogenated →



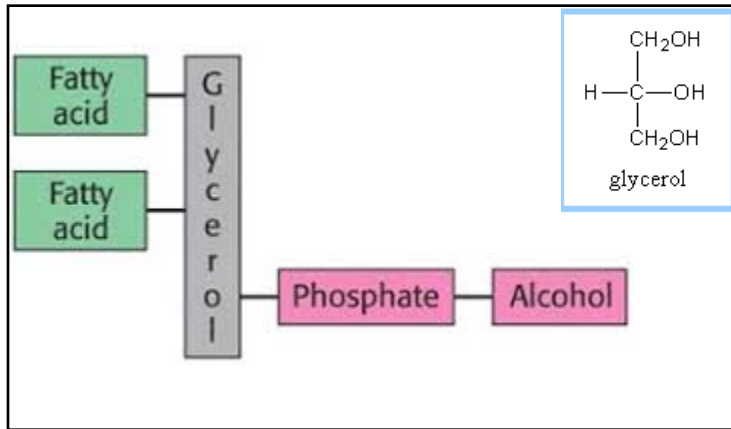
body tries to use TRANS/SATURATED form which is optimal for CIS.

ALTER THE RIGILIGY →

BAD FOR HEALTH (ADELTARATION OF FOOD).

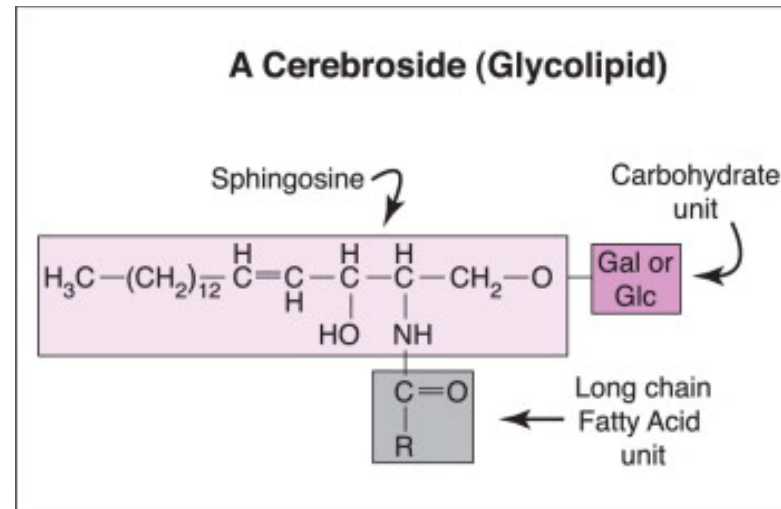
Three major kind of membrane lipids

➤ Phospholipids



A phospholipid (four components):
Glycerol backbone + fatty acids
+ a phosphate + alcohol +

➤ Glycolipids



Sugar containing lipids (three
Component): Sphingosine + Sugar+Fatty acid

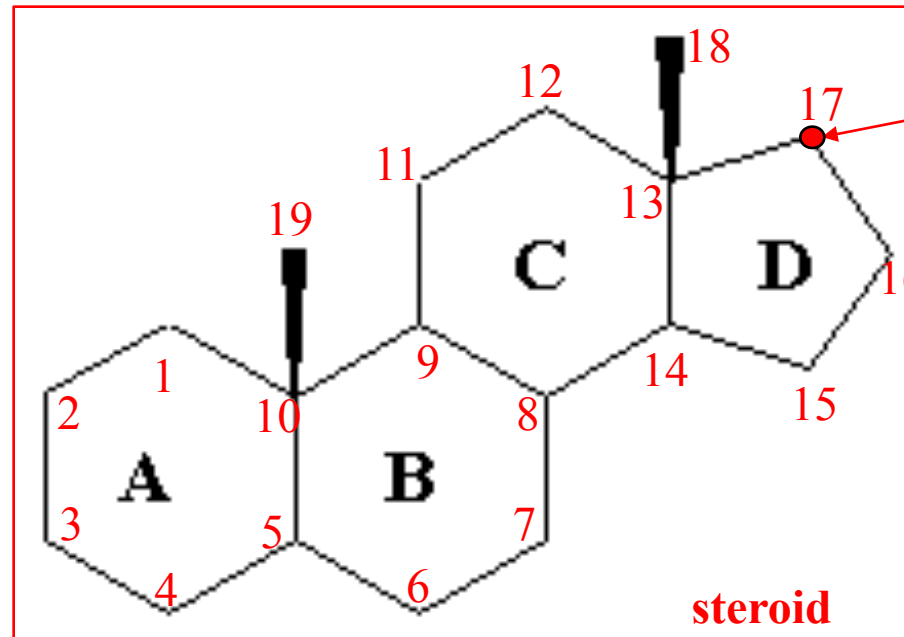
➤ Cholesterol

3. Steroid



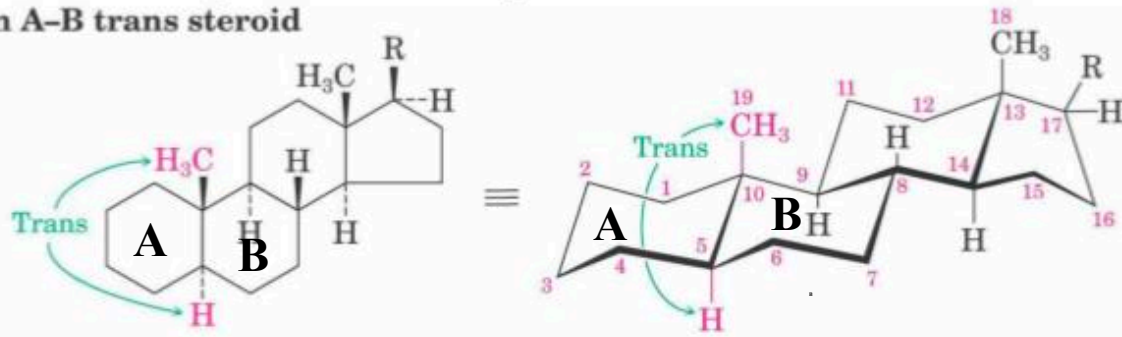
“Anabolic steroids” : **synthetic derivatives of testosterone**

A steroid is a biologically active organic compound with **four rings (A, B, C, D)** arranged in a specific molecular configuration

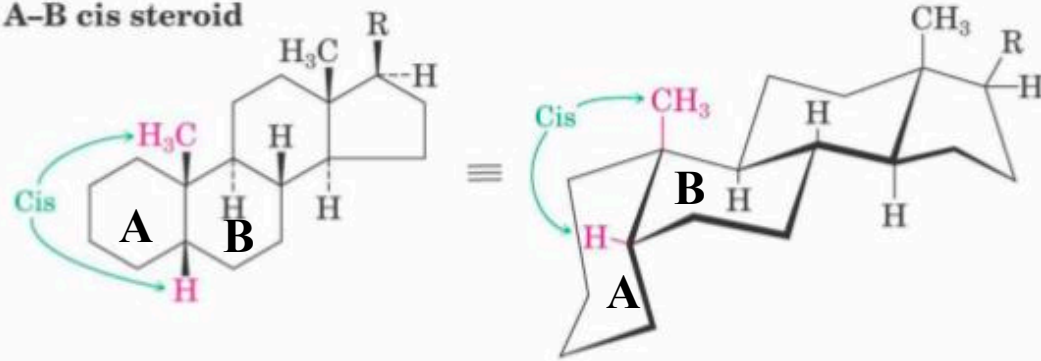


Various substituent at C17 generate different Steroids

An A-B trans steroid

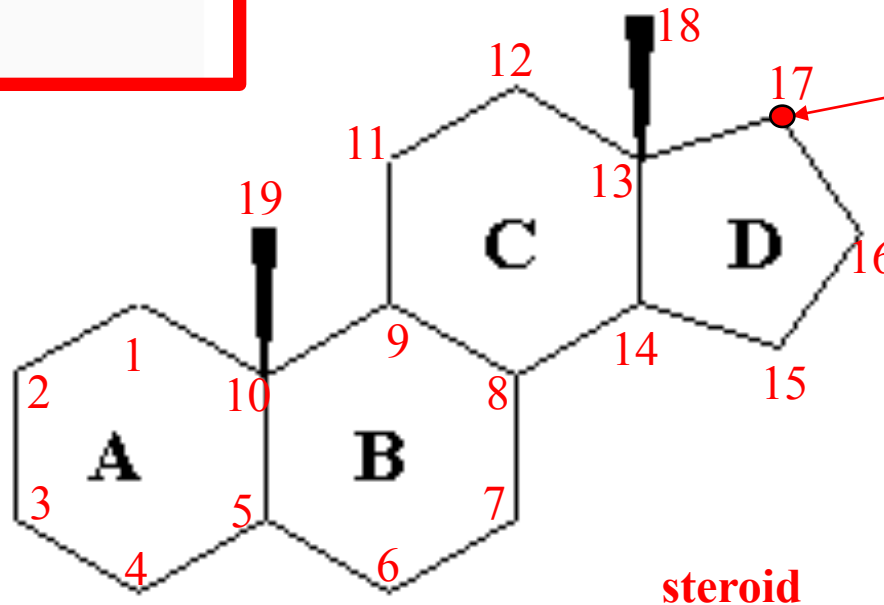


An A-B cis steroid



Note (A-B ring):

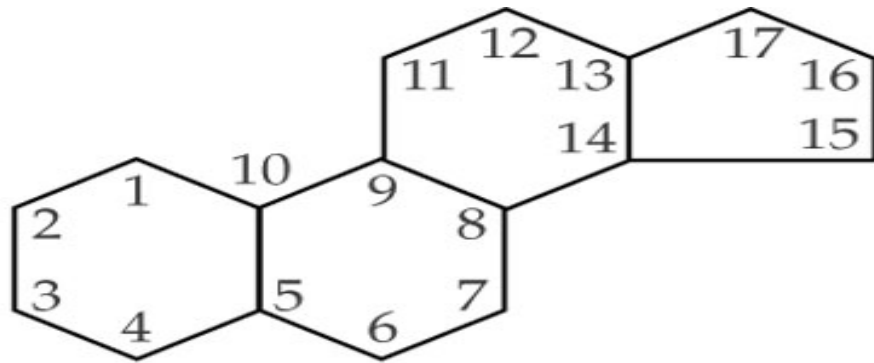
- -CH₃ in carbon 10 is axial (pointing UP)
- Cis and trans steroid based on A-B junction
- (B-C) & C-D are usually trans



Various substituent at C17 generate different Steroids

Steroid refer to molecules that include

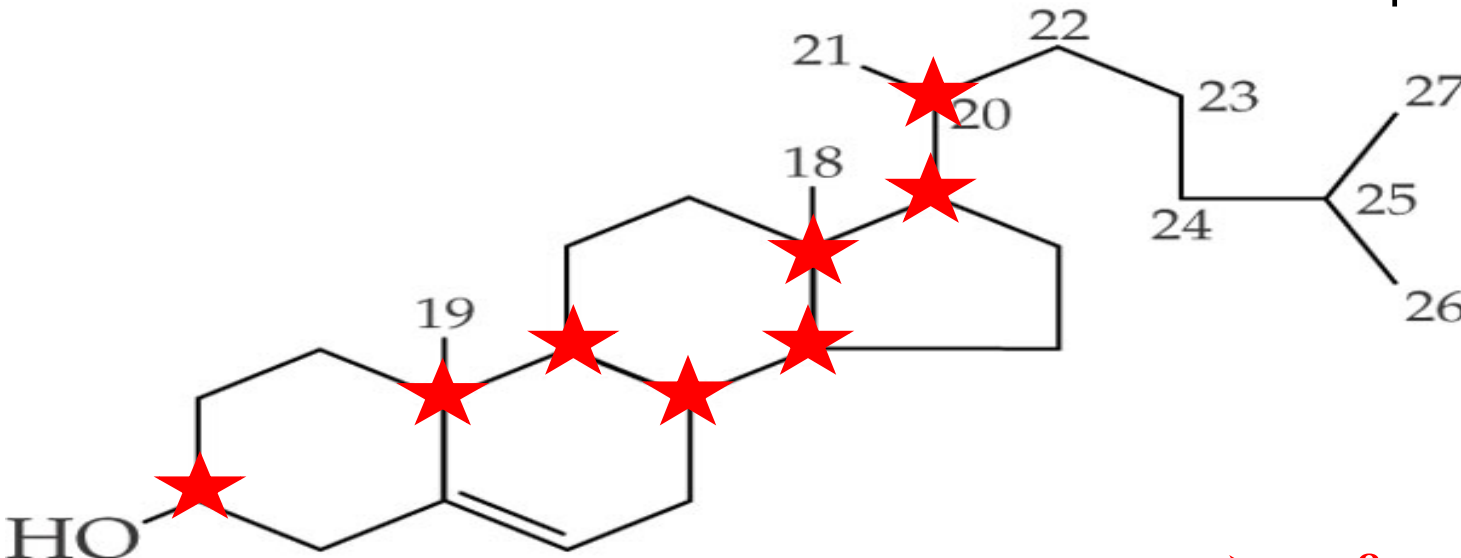
- **Cholesterol**
- **Hormones**
- **Certain vitamins**



Steroid nucleus

Cholesterol

- **Cholesterol :**
- based on steroid nucleus constituent of cell membranes,
(**rigid** ring system + short branched hydrocarbon tail.)
- **hydrophobic.**
- the polar group, a **hydroxyl**, making it **amphipathic**.



Cholesterol

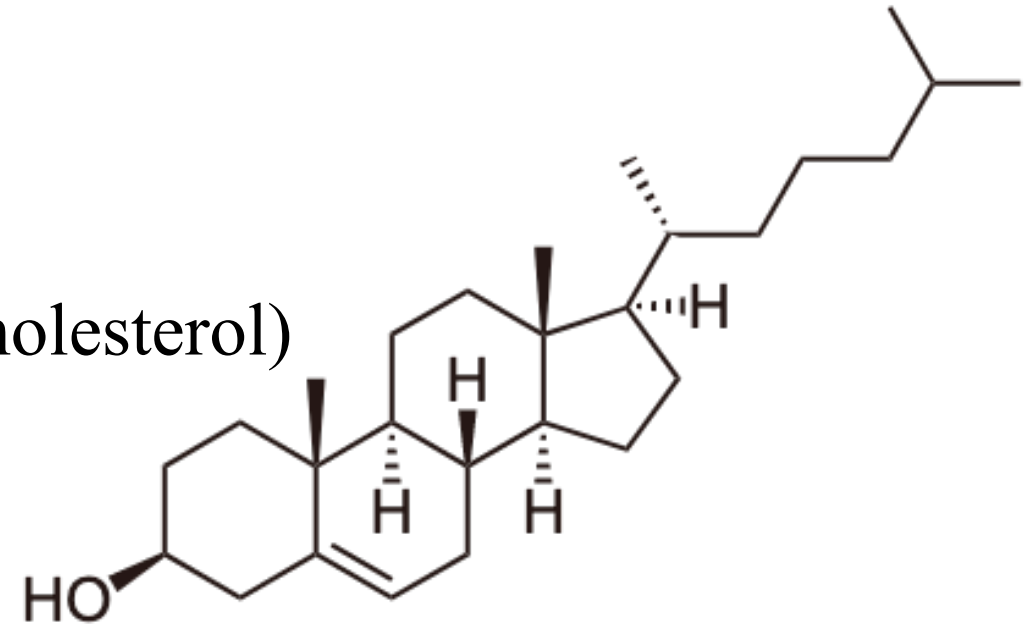
- **$2^8 = 256$ possible stereoisomer**
- **1 of 256 is cholesterol**

Cholesterol

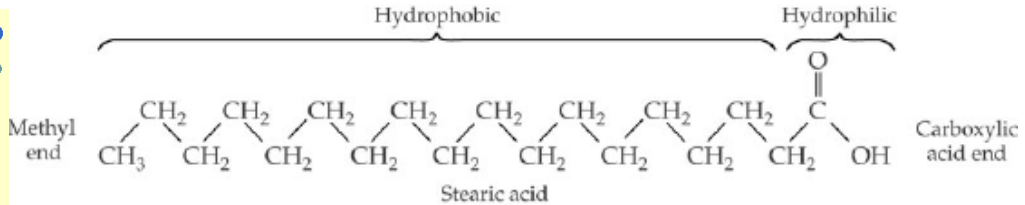
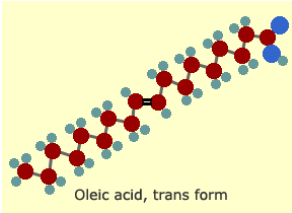
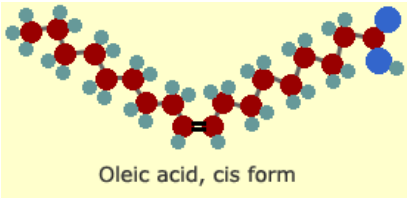
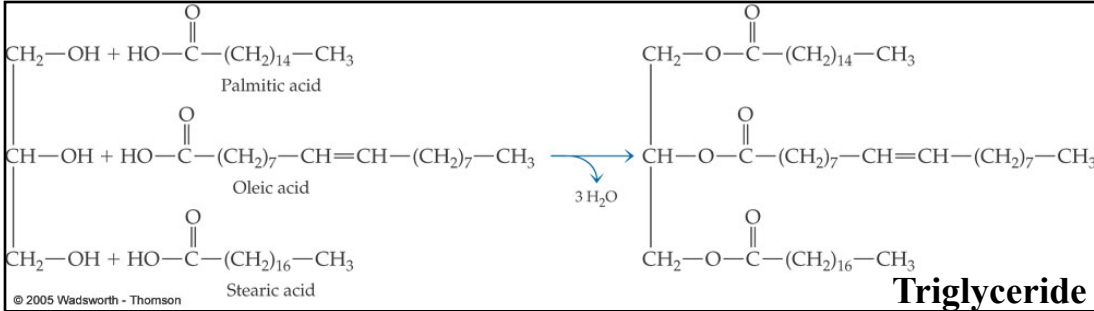
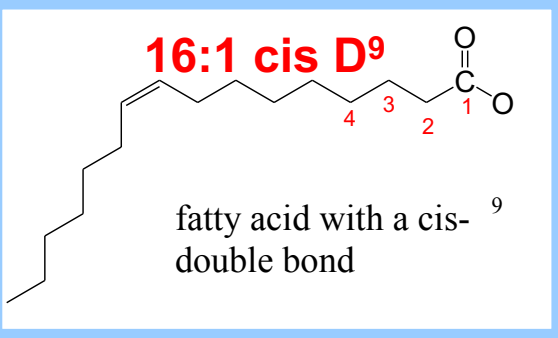
1. Almost all cholesterol in our body is made by our body. NOT from our Diet.
2. Body makes 3000 mg/day

Why?

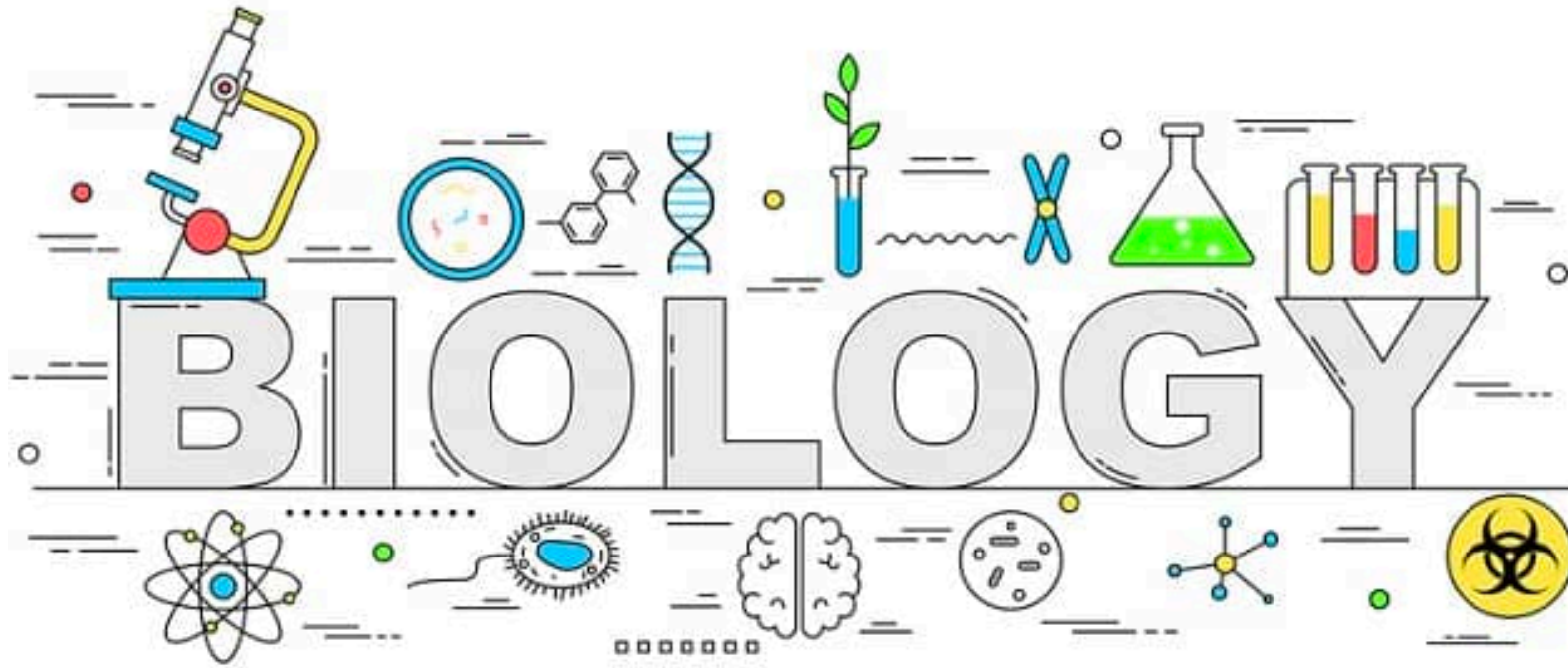
1. Cell membranes (Brain: lots and lots of cholesterol)
2. BILE production
3. Vitamin D, A, K2, E
4. Hormone
5. Myelin synthesis (lipid layer around nervous system)
6. Helps immune system (control WBC)
7. Bind and neutralize toxin from bacteria



Take home



Bad (increases the rigidity)



Next: Lipid transport